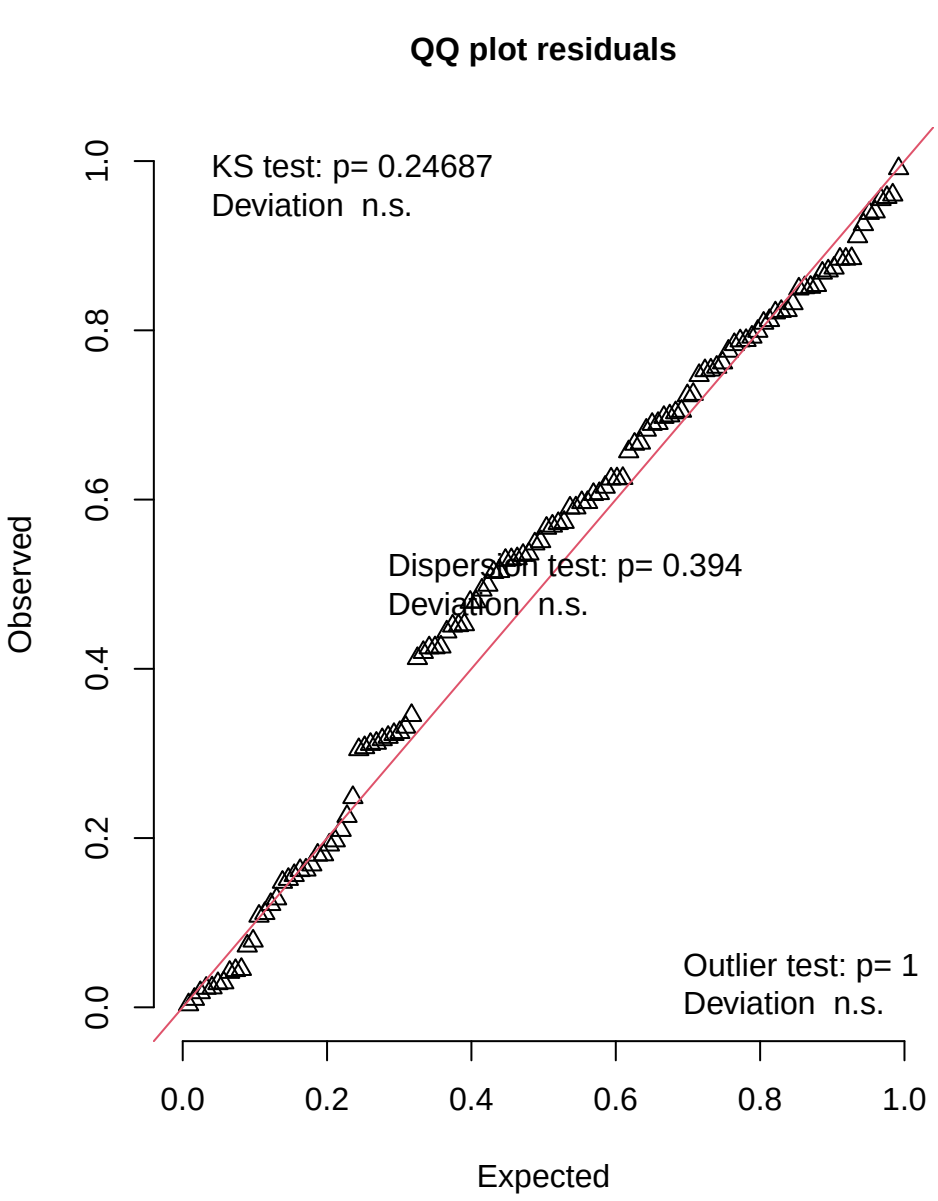
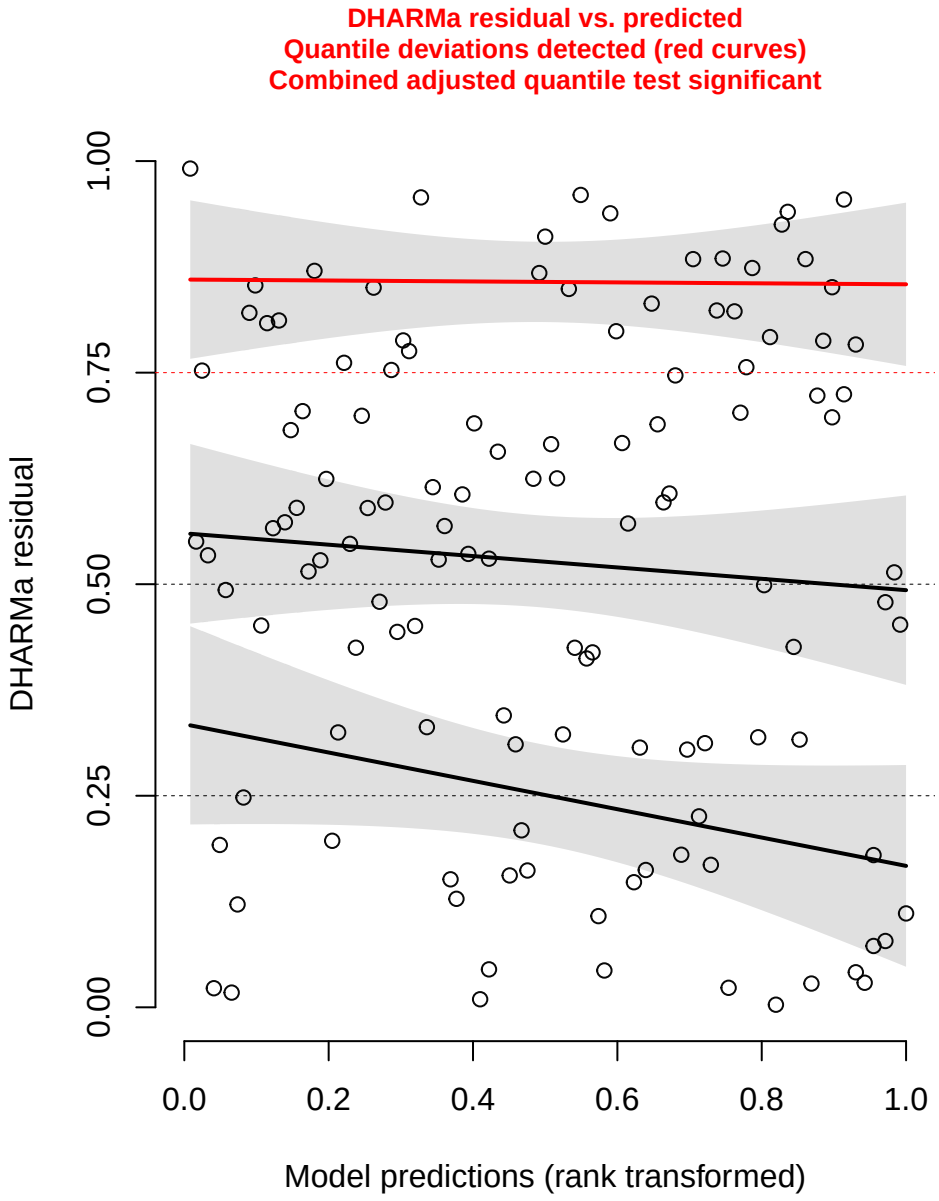
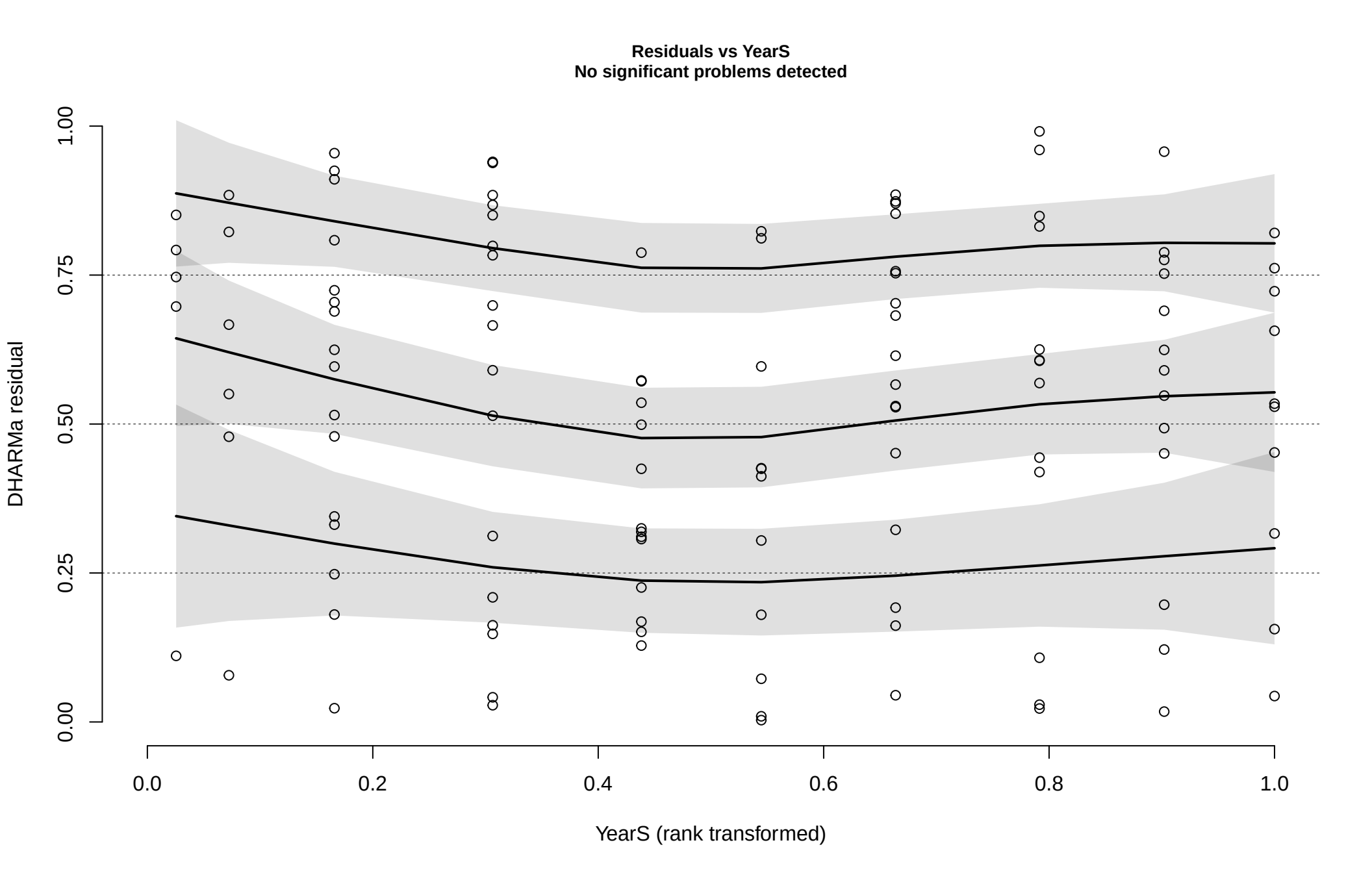
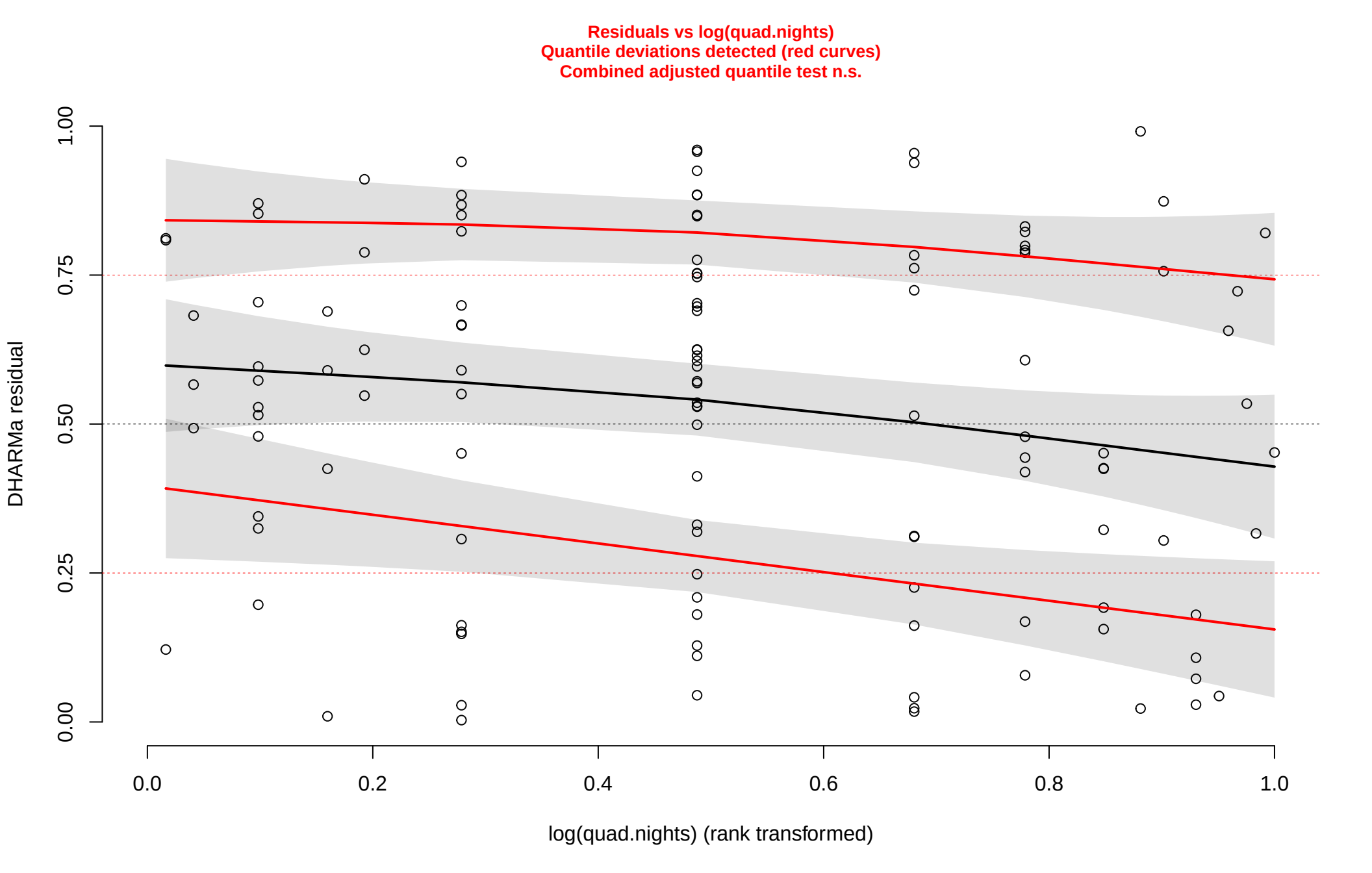
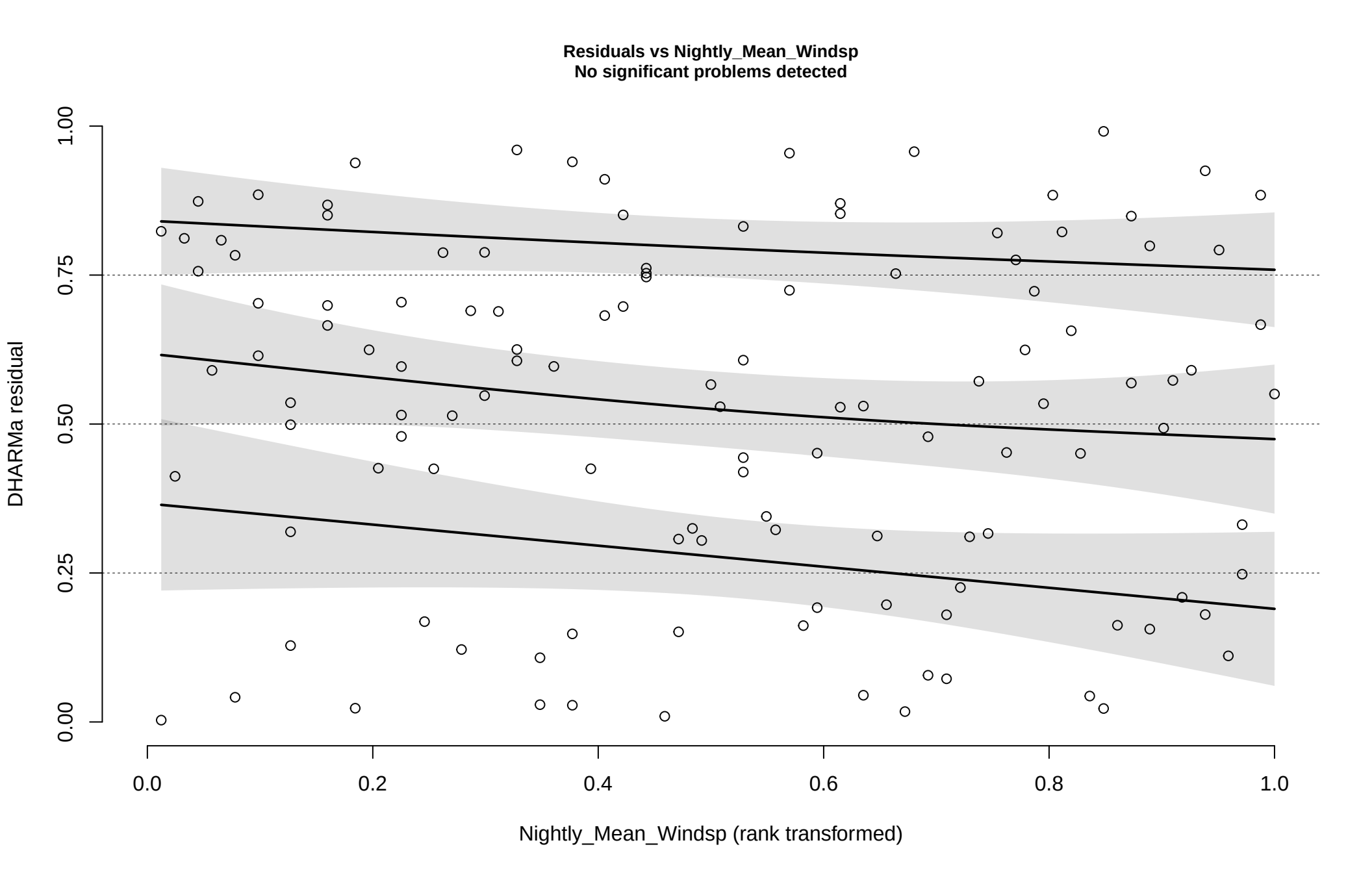
DHARMa residuals: Northern Region_MYYU

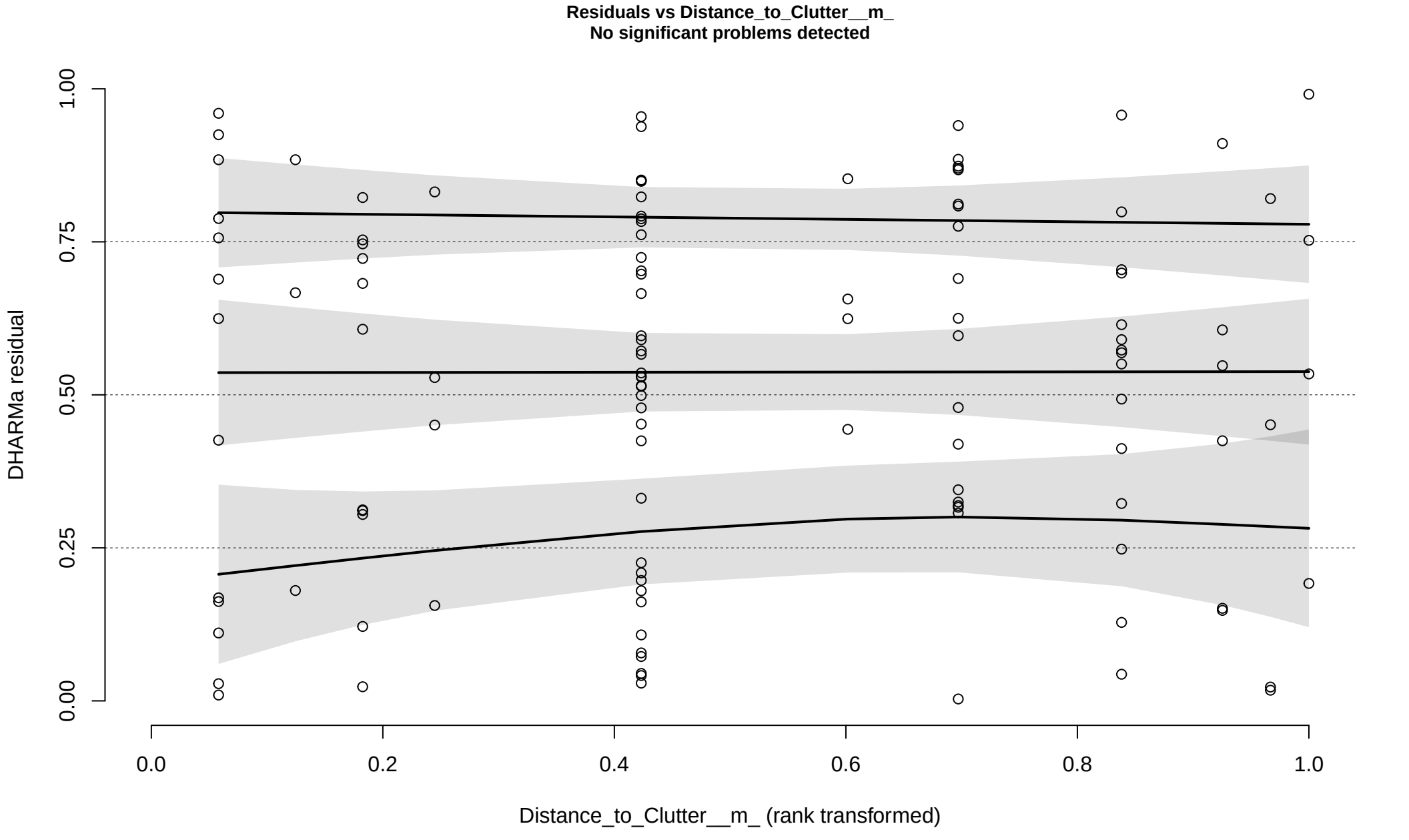


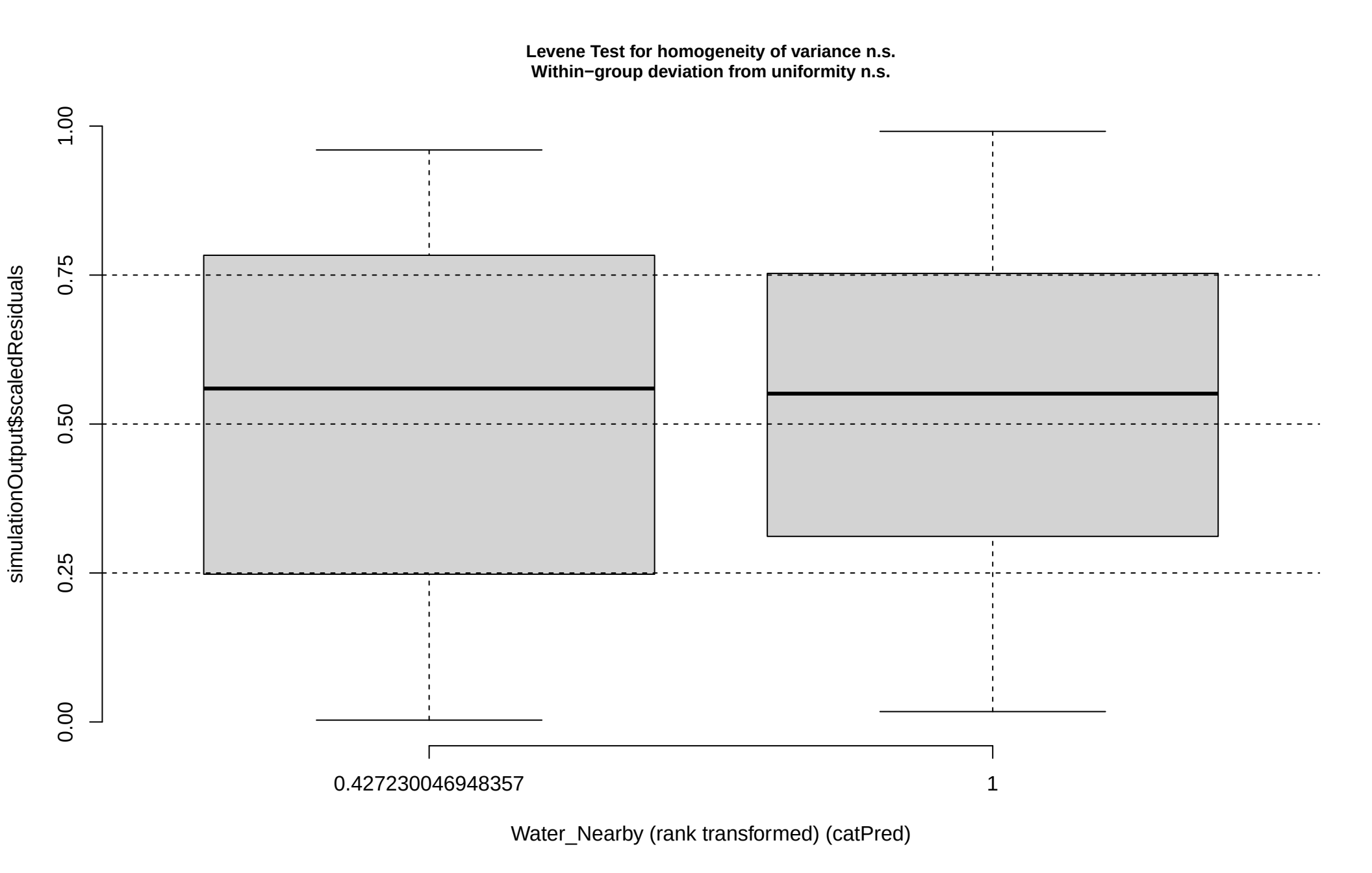


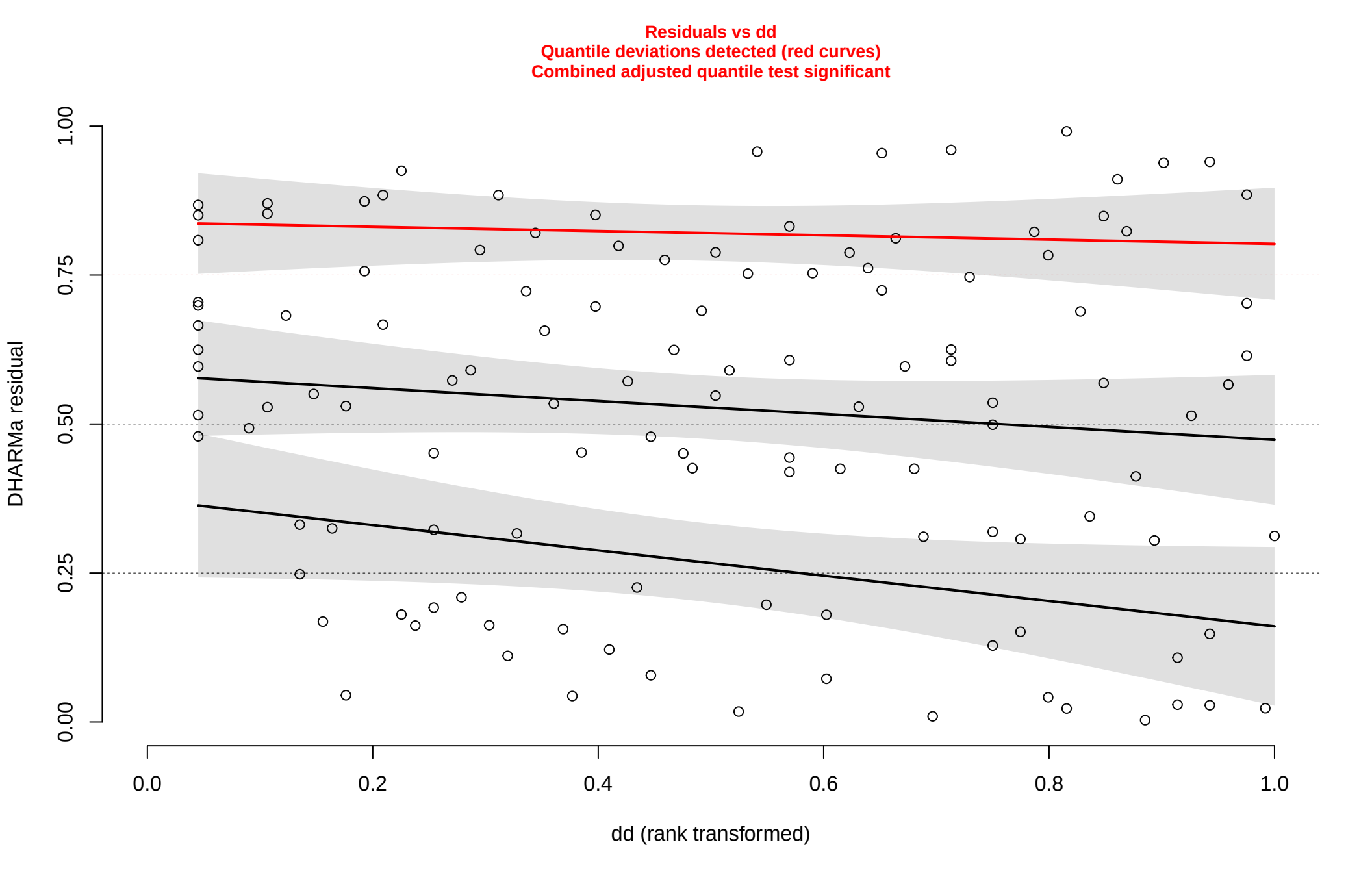

Residuals vs YearS
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

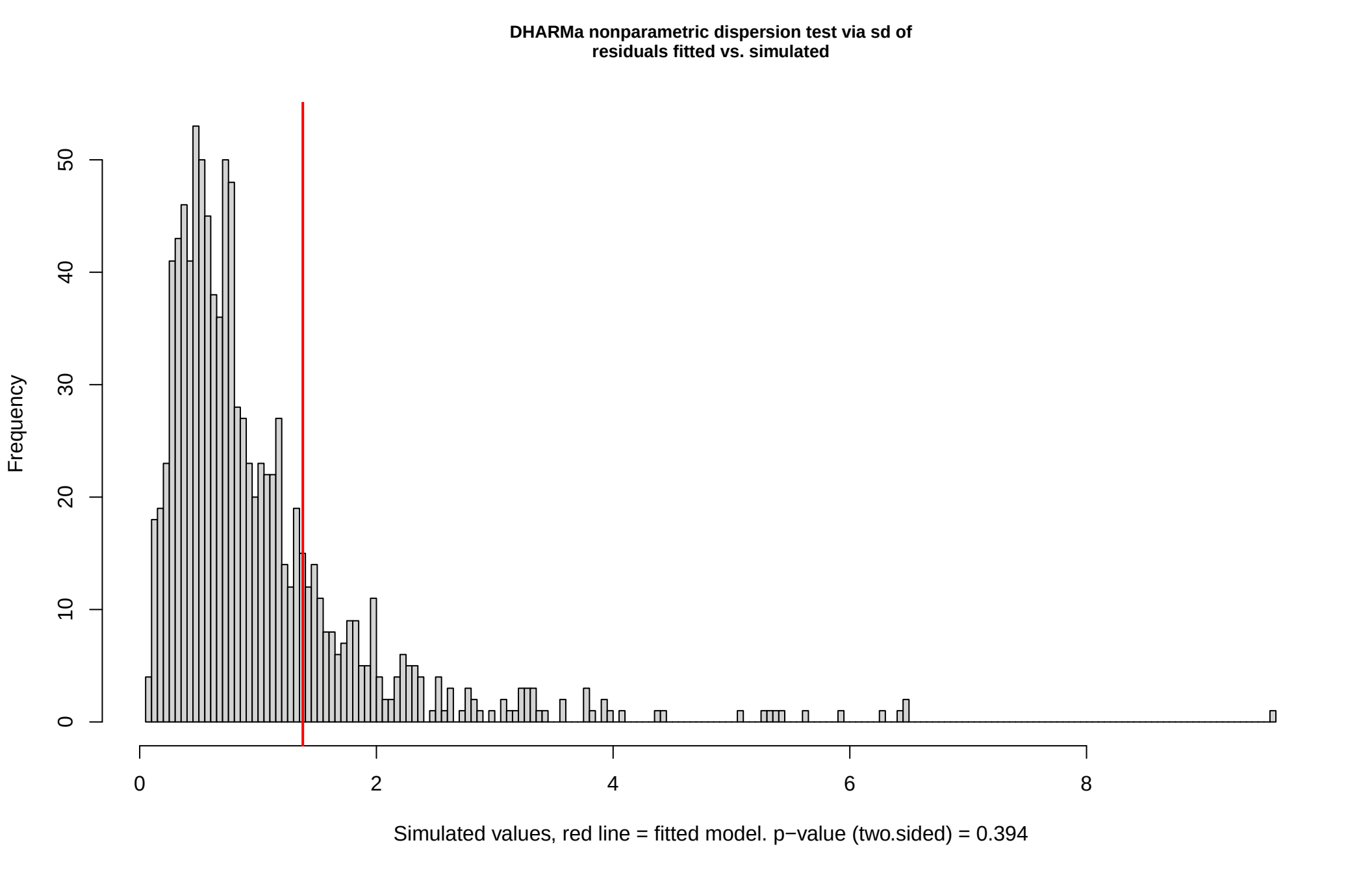

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

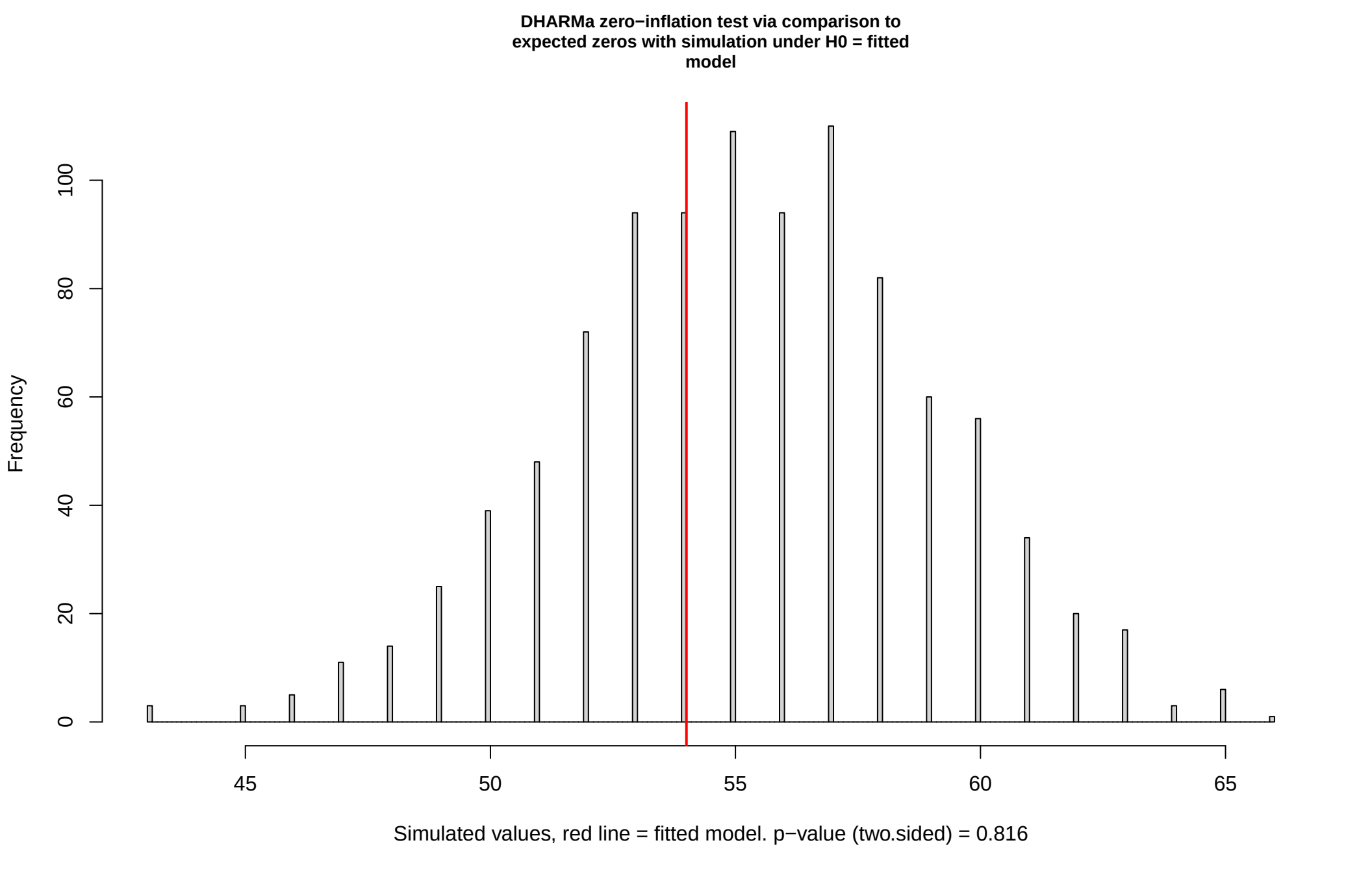

Residuals vs Nightly_Mean_Windsp
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Windsp (rank transformed)

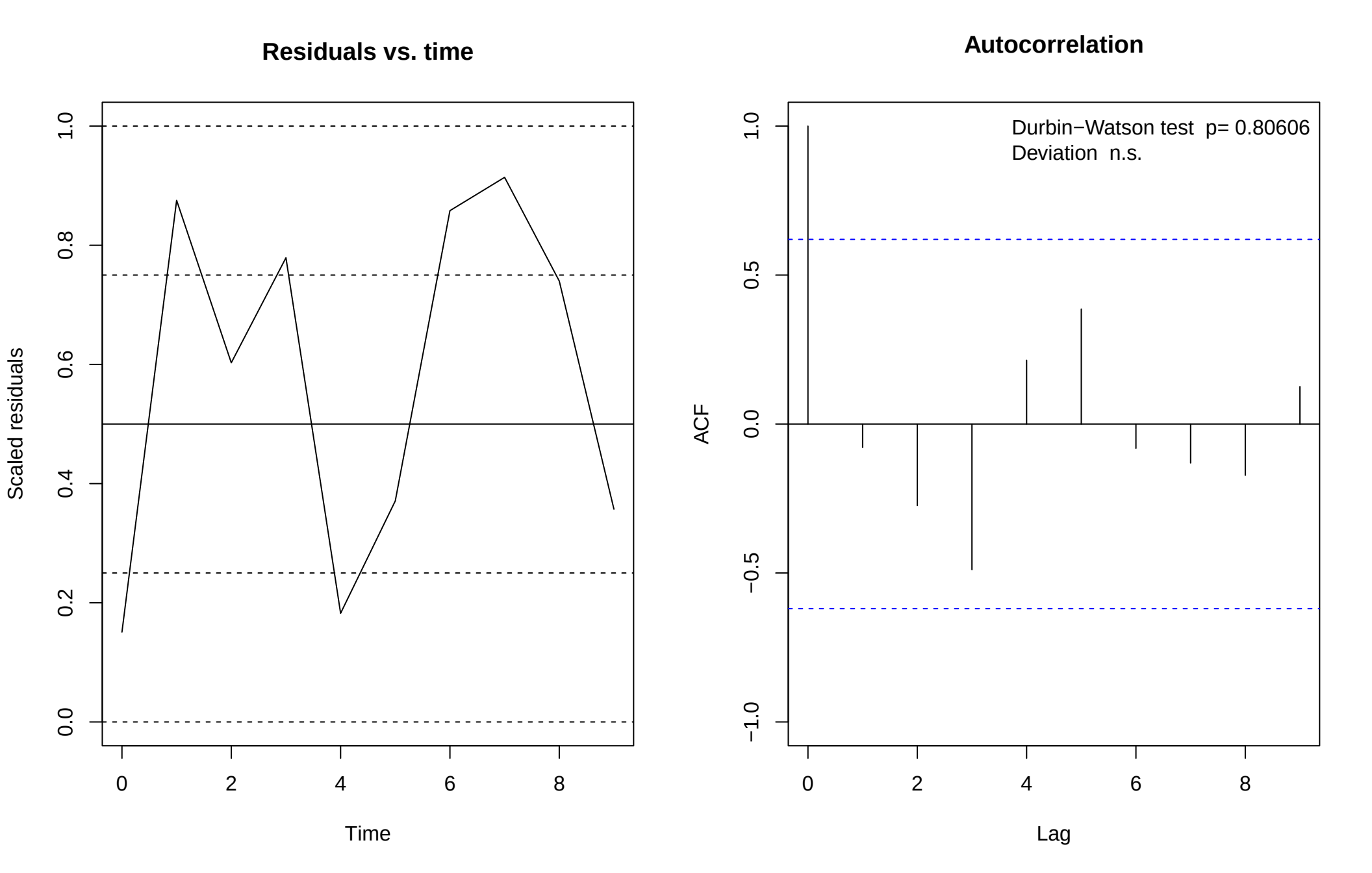

Residuals vs Distance_to_Clutter__m_
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

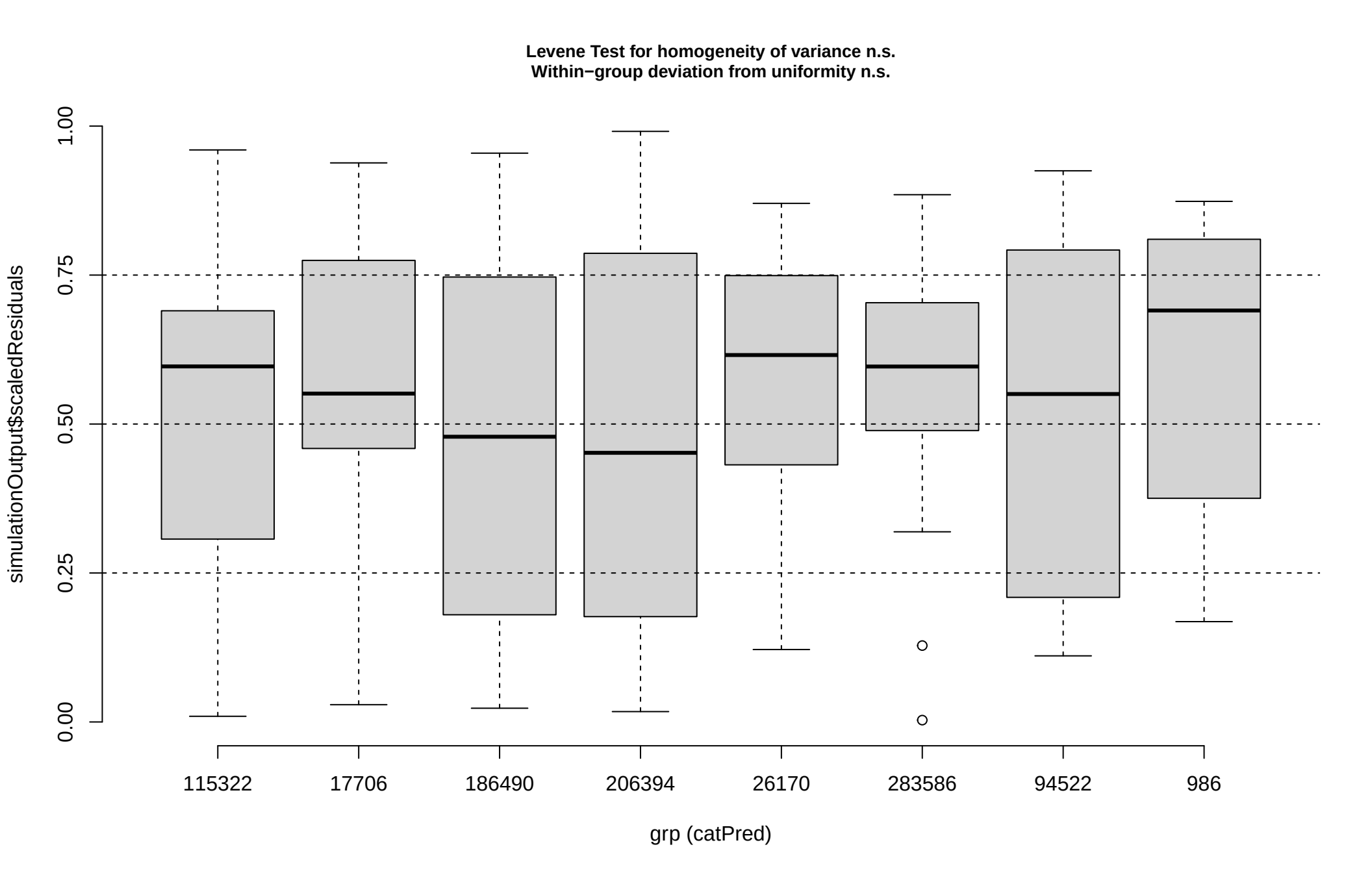

Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
0.427230046948357
1
Water_Nearby (rank transformed) (catPred)

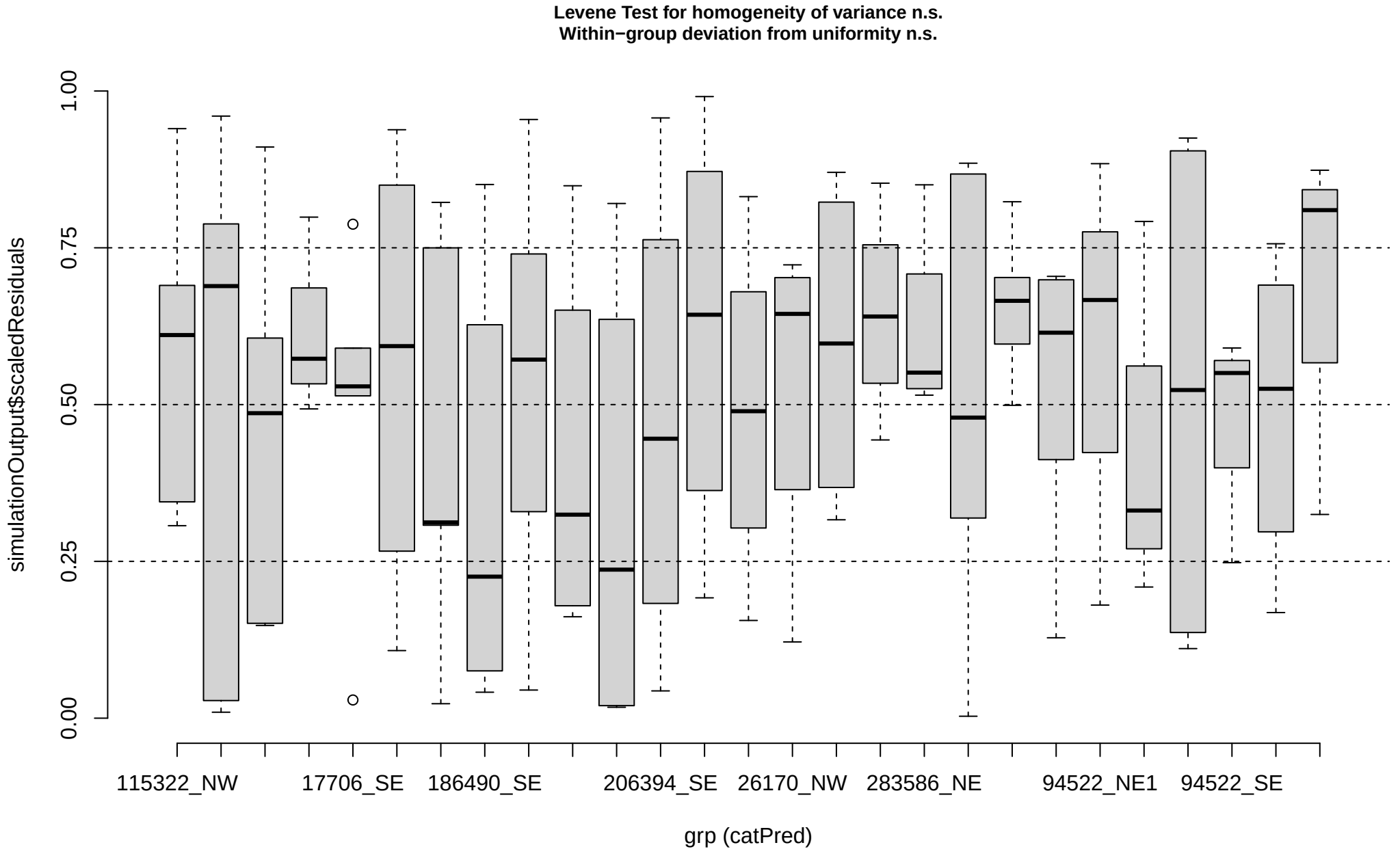

Residuals vs dd
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
dd (rank transformed)

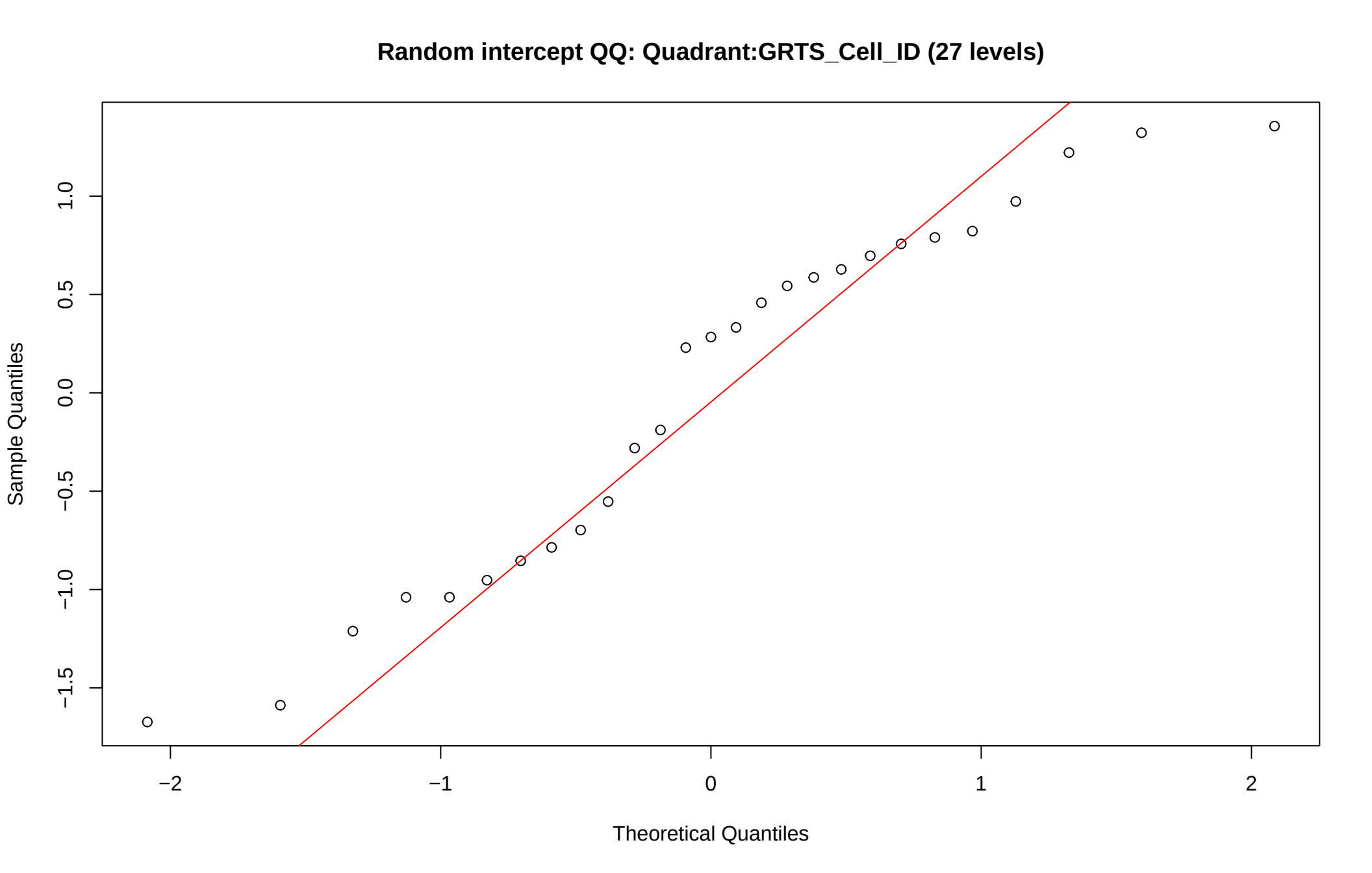

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
10
20
30
40
50
0
2
4
6
8
Simulated values, red line = fitted model. p−value (two.sided) = 0.394

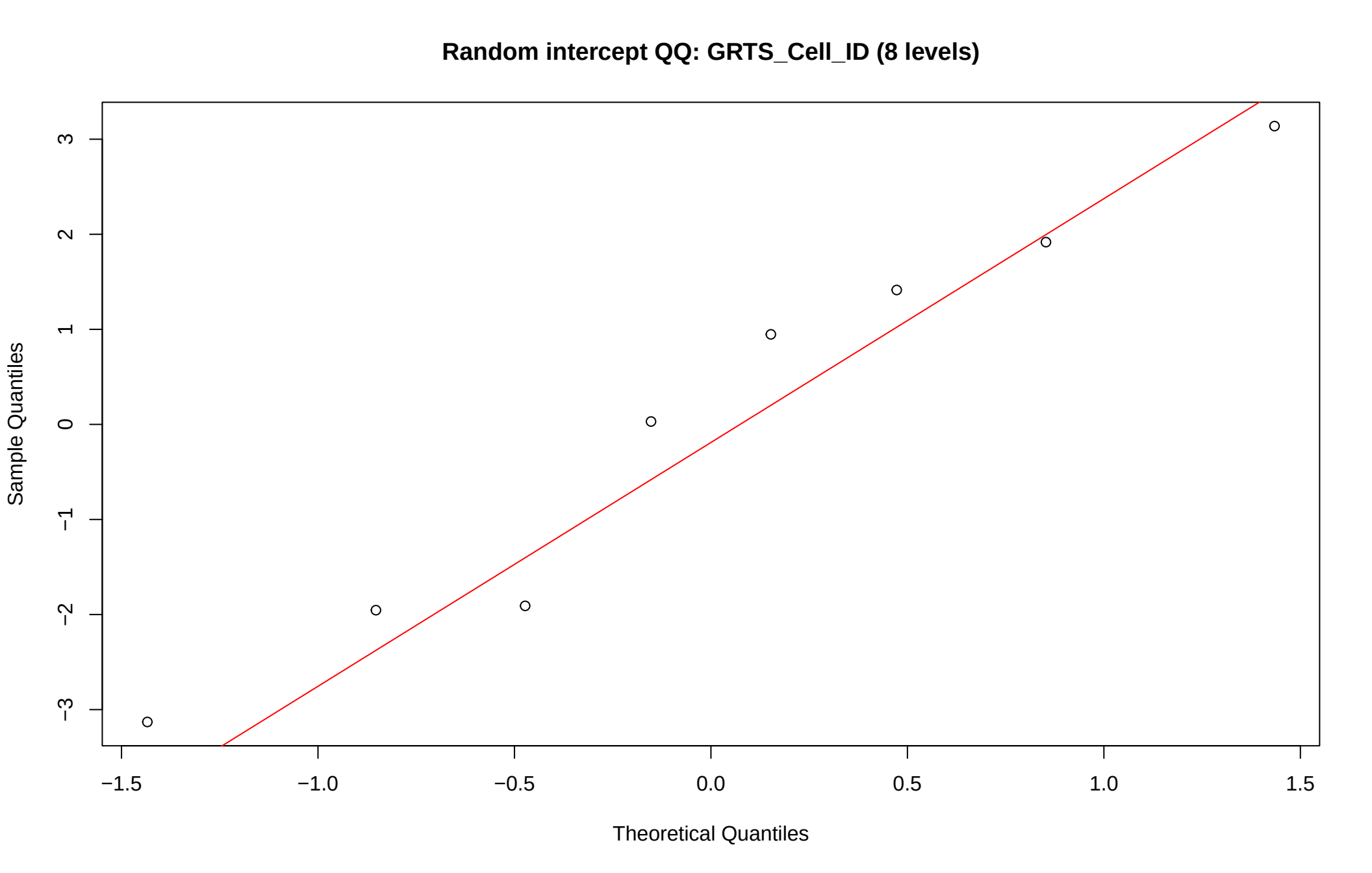

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
20
40
60
80
100
45
50
55
60
65
Simulated values, red line = fitted model. p−value (two.sided) = 0.816


Residuals vs. time
Scaled residuals
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Time
Autocorrelation
ACF
Durbin−Watson test p= 0.80606
Deviation n.s.
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
115322
17706
186490
206394
26170
283586
94522
986
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
115322_NW
17706_SE
186490_SE
206394_SE
26170_NW
283586_NE
94522_NE1
94522_SE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (27 levels)
Sample Quantiles
Theoretical Quantiles
−1.5
−1.0
−0.5
0.0
0.5
1.0
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (8 levels)
Sample Quantiles
Theoretical Quantiles
3
2
1
0
−1
−2
−3
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5